Task 2 — Loan Approval Prediction

Alfido Tech Data Science Internship | Project deliverable

Task 2: Loan Approval Prediction

Dataset and methodology

• Source: `data/raw/loan_approval_dataset.csv`.

• The latest saved experiment compares Logistic Regression and Decision Tree.

• Metrics below come from `outputs/task2_model_comparison.csv`.

• The results describe this dataset and evaluation split only.

Latest model comparison

• **Logistic Regression** — accuracy: 0.9356 (93.56%); ROC-AUC (one-vs-rest): 0.9748.

• **Decision Tree** — accuracy: 0.9778 (97.78%); ROC-AUC (one-vs-rest): 0.9938.

• Highest ROC-AUC in this comparison: Decision Tree (0.9938). This does not by itself establish that the model is suitable for real lending decisions.

Interpretation and limitations

• Compare precision, recall, F1-score and confusion-matrix errors before choosing a model.

• Check for target leakage, preprocessing leakage, class imbalance and appropriate stratification.

• Assess calibration, subgroup fairness and the costs of false approvals and false rejections.

• These educational results are not a credit decision system and do not demonstrate real-world lending suitability.

Outputs

• `outputs/task2_model_comparison.csv`

• `outputs/task2_logistic_regression_report.txt`

• `outputs/task2_decision_tree_report.txt`

• `outputs/task2_logistic_confusion_matrix.png`

Recommendations and next steps

• Compare logistic regression with a tree-based model using the same train/test split.

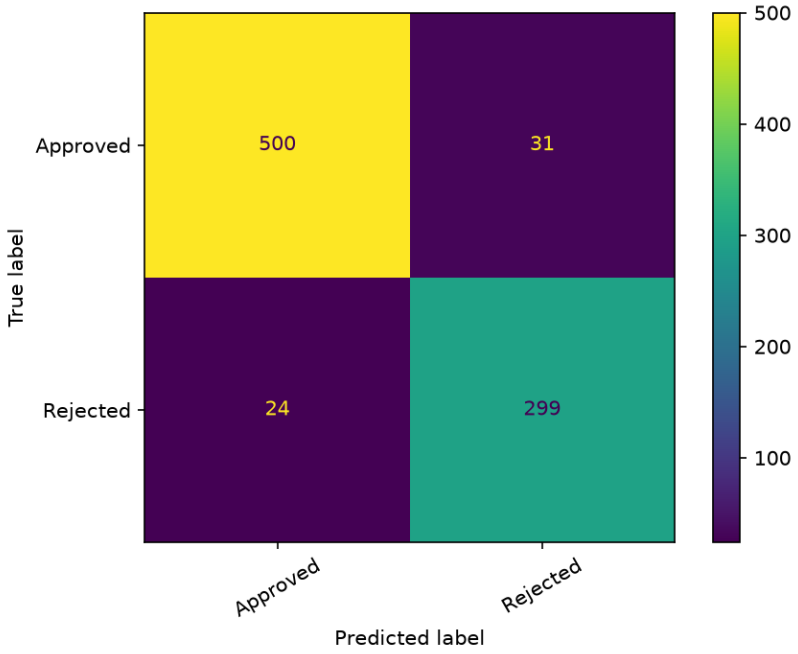
• Report precision, recall, F1-score and ROC-AUC, not accuracy alone.

• Review class imbalance and confusion-matrix errors, especially false approvals and false rejections.

• Choose any deployment threshold only after validation, calibration and cost analysis.

• Treat results as educational dataset performance, not as a real lending decision or guarantee.

Task2 Logistic Confusion Matrix


500
31
24
299
Approved
Rejected
True label
Approved
Rejected
Predicted label
500
400
300
200
100